

Diets and hormonal levels in postmenopausal women with or without breast cancer.

The role of diet in breast cancer (BC) risk is unclear. Fiber could reduce BC risk, through the enterohepatic circulation of estrogens. We examined the relationship between diet and sex hormones in postmenopausal women with or without BC. Thirty-one postmenopausal women (10 omnivores, 11 vegetarians, and 10 BC omnivores) were recruited. Dietary records (5 days) and hormone levels (3 days) were evaluated on 4 occasions over 1 yr. Vegetarians showed a lower fat/fiber ratio, a higher intake of total and cereal fiber (g/d)/body weight (kg), a significantly lower level of plasma estrone-sulfate, estradiol, free-estradiol, free-testosterone, and ring D oxygenated estrogens, and a significantly higher level of sex-hormone-binding-globulin than BC subjects. Fiber was consumed in slightly larger amounts by omnivores than by BC subjects. Omnivores had significantly lower plasma testosterone and estrone-sulfate but higher sex-hormone-binding-globulin than BC subjects. No difference was found for the urinary 16-oxygenated estrogens. However, the 2-MeO-E1/2-OH-E1 ratio was significantly lower in omnivores than in BC group. This ratio is positively associated with the fat/fiber ratio. In conclusion, testosterone may contribute to causing alterations in the levels of catechol estrogens and 16-oxygenated estrogens. The fat/fiber ratio appears to be useful in evaluating dietary effects on estrogen metabolism.